

## Laiton Hedley

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My name is Laiton Hedley, I am a Lecturer and late-stage PhD candidate in Psychological Science at the University of Newcastle (Australia). My research examines human-human and human-machine coordination in dynamic task environments, with a focus on team performance, division of labour, real-time behaviour, and computational approaches to understanding interactive cognition. I have experience designing and delivering online and laboratory experiments, analysing large behavioural and time-series datasets, and communicating research through publications, conference presentations, and teaching.

### EDUCATIONAL BACKGROUND:

#### PhD Psychological Science

University of Newcastle

Estimated Completion October 2026

#### Bachelor of Psychological Science: First Class Honours

University of Newcastle

Completed 2021

#### Diploma of Music

Australian Institute of Music

Completed 2016

### WORK EXPERIENCE:

#### Software Developer

jamovi (open-source statistical software)

December 2025 – Current

Contributor to the jamovi project; an open-source statistical software for researchers and students. I am developing modules specifically for data visualisation and creating formal documentation of jamovi functions.

- Developed modules for data visualisation and analyses.
- Created documentation for specific data handling functions.
- Presented products to key stakeholders.

#### Lecturer Level A (University of Newcastle)

School of Psychological Sciences (University of Newcastle)

March 2023 – Current

A tertiary education setting where I teach undergraduate Psychology students Survey Design, Research Methods, Data Handling Practices, and Data Analyses.

- Created learning materials for students.
- Taught survey design, analysis skills, and research methods.
- Conducted student evaluation and marking.

#### PhD Student (Paid Position)

The Cognition Laboratory (University of Newcastle)

March 2022 – Current

My PhD project involves literature review, experimental design, deploying experiments in lab and online, data analysis, modelling of humans and human-AI behaviour, writing and publishing findings.

- Created MATLAB and R scripts for data handling, visualisation, and analyses.
- Cleaned and analysed large time-series data sets.
- Conducted literature reviews, summarised findings, and synthesised work for publication.

#### Defence Internship (Paid Internship)

Alpha Echo: Defence Contractor (Newcastle)

November 2023 – April 2024

Through my PhD I was able to secure 3-months of funding to work alongside Alpha Echo whereby I helped develop evidence-based products and business proposals for a local Cybersecurity company that contracts with the Australian Defence force.

- Develop evidence-based cybersecurity products.
- Conducted a literature review on Human touch points in cybersecurity.
- Presented products to key stakeholders.

#### Honours Student & Research Assistant (Paid Internship)

Cognition Laboratory (University of Newcastle)

December 2019 – November 2021

A learning experience regarding cognitive modelling guided by an academic supervisor (Prof Ami Eidels), where I participated in one-on-one learning sessions and was provided coding tasks to complete.

- Created MATLAB scripts representing human decision-making.
- Produced written summaries of codes and minutes of meetings.
- Developed coding skills useful for data analysis and cognitive modelling.

#### Research Assistant

Functional Neuroimaging Lab (University of Newcastle)

September 2019 – November 2019

A lab setting conducting electroencephalogram (or EEG) experiments done in pairs (myself and the PhD students) with one participant at a time.

- Collected and stored data.
- Applied interpersonal skills to ensure participants are comfortable in the lab.
- Followed detailed guidelines for setting up EEG caps, setting up experiment and cleaning EEG caps.

#### RESEARCH AND SOFTWARE SKILLS:

- Experimental design & implementation
- Behavioural measurement
- Participant management (interviews & recruitment)
- Ethics compliance
- Python (experimental programming)
- Microsoft Excel (basic data handling)
- JaMovi/JASP/SPSS (statistical analyses)
- OneDrive (data storage)
- Overleaf and LaTeX (for scientific communication)
- MATLAB, Python, & R (data manipulation)

#### REFEREES:

**Dr Ami Eidels** (primary PhD supervisor)  
Professor (The University of Newcastle)

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[ami.eidels@newcastle.edu.au](mailto:ami.eidels@newcastle.edu.au)

**Dr Scott Brown** (co-supervisor of PhD)  
Professor (The University of Newcastle)

+61 423 176 112 (Australia)  
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**Dr Murray Bennett** (frequent collaborator)  
Post-doctoral researcher (Ohio State University)

+1 726 209 1036 (United States)  
[bennett.1755@osu.edu](mailto:bennett.1755@osu.edu)

#### RESEARCH OUTPUT:

##### Publications:

Hedley, L, Russel, L., Bennet, M. S., & Eidels, A. (accepted). A Novel Simulation Benchmark for Team Performance: Benchmarking a Benchmark. *Chinese Journal of Psychology*.

Hedley, L. G, Bennett, M. S, Love, J., Houpt, J., Brown, S., & Eidels, A. (2023). The Relationship Between Teaming Behaviours and Joint Capacity of Hybrid Human-Machine Teams. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45.  
<https://escholarship.org/uc/item/61b5c7gv>

Bennett, M. S., Hedley, L., Love, J., Houpt, J. W., Brown, S. D., & Eidels, A. (2022). Human Performance in Competitive and Collaborative Human-Machine Teams. *Topics in Cognitive Science*. <https://doi.org/10.1111/tops.12683>

Hedley, L, Bennet, M. S., Russel, L., Love, J., Eidels, A., & Brown, S. Division of Labour and Team Performance: Human-Human Teams compared to Hybrid Human-Machine Teams. (under review). <https://doi.org/10.31234/osf.io/s32yc>

Fitch, A. K., Bennett, M. S., Hedley, L., & Kvam, P. D. (in-prep). *Real-time cognitive modeling using simulation-inversion networks*. Manuscript under review at *British Journal of Mathematical Psychology*.

##### Conference Presentations:

Experimental Psychology Conference (EPC: 2023): The Relationship Between Teaming Behaviours and Joint Capacity of Hybrid Human-Machine Teams

Australian Mathematical Psychology Conference (AMPC: 2023): The Relationship Between Teaming Behaviours and Joint Capacity of Hybrid Human-Machine Teams

Cognitive Science Society Conference (2023): The Relationship Between Teaming Behaviours and Joint Capacity of Hybrid Human-Machine Teams

Society of Australasian Social Psychology Conference (2023): The Relationship Between Teaming Behaviours and Joint Capacity of Hybrid Human-Machine Teams

Performance and Expertise Research Centre Conference (PERCC: 2023): The Relationship Between Teaming Behaviours and Team Performance of Hybrid Human-Machine Teams

PERCC: (2024): Division of Labour and Team Performance: Human-Human Teams compared to Hybrid Human-Machine Teams

AMPC (2025): Division of Labour and Team Performance: Human-Human Teams compared to Hybrid Human-Machine Teams

EPC (2025): Division of Labour and Team Performance: Human-Human Teams compared to Hybrid Human-Machine Teams

Human-AI Complementarity Workshop (2025): Division of Labour and Team Performance: Human-Human Teams compared to Hybrid Human-Machine Teams